

附录 A 原理性公式汇总

A.1 CIE 标准观察者函数与波长轴

$$\lambda = 380, 385, \dots, 780, \quad \Delta\lambda = 5 \text{ nm}$$

$$\bar{x}(\lambda), \quad \bar{y}(\lambda), \quad \bar{z}(\lambda)$$

其中, $\bar{x}(\lambda)$ 、 $\bar{y}(\lambda)$ 、 $\bar{z}(\lambda)$ 为 CIE 1931 2° 标准观察者函数, 是光谱转换为 XYZ 的人眼响应权重。

A.2 光谱到 XYZ、xy 与光通量

$$X = \sum S(\lambda)\bar{x}(\lambda)\Delta\lambda, \quad Y = \sum S(\lambda)\bar{y}(\lambda)\Delta\lambda, \quad Z = \sum S(\lambda)\bar{z}(\lambda)\Delta\lambda$$

$$x = \frac{X}{X+Y+Z}, \quad y = \frac{Y}{X+Y+Z}, \quad \Phi_v = 683Y$$

A.3 xyY 到 XYZ

$$X_c = \frac{x_c}{y_c}Y_c, \quad Y_c = Y_c, \quad Z_c = \frac{1 - x_c - y_c}{y_c}Y_c, \quad c \in \{R, G, B, T\}$$

A.4 RGB 权重求解

$$A = \begin{bmatrix} X_R & X_G & X_B \\ Y_R & Y_G & Y_B \\ Z_R & Z_G & Z_B \end{bmatrix}, \quad \begin{bmatrix} r \\ g \\ b \end{bmatrix} = A^{-1} \begin{bmatrix} X_T \\ Y_T \\ Z_T \end{bmatrix}$$

$$r = 0.640411080154, \quad g = 0.882500416468, \quad b = 0.265311206990$$

A.5 通道光谱亮度校准

$$\alpha_c = \frac{Y_c}{683 \sum S_c^{raw}(\lambda) \bar{y}(\lambda) \Delta\lambda}, \quad S_c(\lambda) = \alpha_c S_c^{raw}(\lambda), \quad c \in \{R, G, B\}$$

$$Y_R = 1.02 \text{ lm}, \quad Y_G = 1.8628 \text{ lm}, \quad Y_B = 0.35 \text{ lm}$$

A.6 RGB 白光合成与目标亮度归一化

$$S_{RGB0}(\lambda) = rS_R(\lambda) + gS_G(\lambda) + bS_B(\lambda)$$

$$S_{RGB0}(\lambda) = 0.640411080154S_R(\lambda) + 0.882500416468S_G(\lambda) + 0.265311206990S_B(\lambda)$$

$$\alpha_T = \frac{2.39}{683 \sum S_{RGB0}(\lambda) \bar{y}(\lambda) \Delta\lambda}, \quad S_{RGB}(\lambda) = \alpha_T S_{RGB0}(\lambda)$$

A.7 单白灯与 RGB 白光混合

$$S_{SingleNorm}(\lambda) \rightarrow 2.39 \text{ lm}, \quad S_{RGBNorm}(\lambda) \rightarrow 2.39 \text{ lm}$$

$$S_{mix}(\lambda, p) = pS_{SingleNorm}(\lambda) + (1 - p)S_{RGBNorm}(\lambda), \quad p = 0, 0.001, \dots, 1$$

A.8 CCT、Duv 与参考光源

$$u = \frac{4X}{X + 15Y + 3Z}, \quad v = \frac{6Y}{X + 15Y + 3Z}$$

$$CCT = \arg \min_T [(u_{test} - u_{BB}(T))^2 + (v_{test} - v_{BB}(T))^2]$$

$$Duv = \sqrt{(u_{test} - u_{BB}(CCT))^2 + (v_{test} - v_{BB}(CCT))^2}$$

$$CCT < 5000K : S_{ref}(\lambda) = S_{BB}(\lambda, T), \quad CCT \geq 5000K : S_{ref}(\lambda) = S_D(\lambda, T)$$

A.9 黑体参考光谱与日光参考光谱

$$S_{BB}(\lambda, T) = \frac{1}{\lambda^5 \left(e^{\frac{c_2}{\lambda T}} - 1 \right)}$$

$$S_D(\lambda, T) = S_0(\lambda) + M_1 S_1(\lambda) + M_2 S_2(\lambda)$$

$$y_D = -3x_D^2 + 2.870x_D - 0.275$$

$$T \leq 7000K : x_D = 0.244063 + \frac{0.09911 \times 10^3}{T} + \frac{2.9678 \times 10^6}{T^2} - \frac{4.6070 \times 10^9}{T^3}$$

$$T > 7000K : x_D = 0.237040 + \frac{0.24748 \times 10^3}{T} + \frac{1.9018 \times 10^6}{T^2} - \frac{2.0064 \times 10^9}{T^3}$$

$$M_1 = \frac{-1.3515 - 1.7703x_D + 5.9114y_D}{0.0241 + 0.2562x_D - 0.7341y_D}, \quad M_2 = \frac{0.0300 - 31.4424x_D + 30.0717y_D}{0.0241 + 0.2562x_D - 0.7341y_D}$$

A.10 标准色样反射光谱与色样 XYZ

$$S_{i,test}(\lambda, p) = S_{mix}(\lambda, p)\rho_i(\lambda), \quad S_{i,ref}(\lambda) = S_{ref}(\lambda)\rho_i(\lambda), \quad i = 1, 2, \dots, 8$$

$$k_j = \frac{100}{\sum S_j(\lambda)\bar{y}(\lambda)\Delta\lambda}, \quad j \in \{test, ref\}$$

$$[X_{i,j}, Y_{i,j}, Z_{i,j}] = k_j \sum S_j(\lambda)\rho_i(\lambda)[\bar{x}(\lambda), \bar{y}(\lambda), \bar{z}(\lambda)]\Delta\lambda$$

A.11 CIE 1960 uv 与色适应

$$u = \frac{4X}{X + 15Y + 3Z}, \quad v = \frac{6Y}{X + 15Y + 3Z}$$

$$c = \frac{4 - u - 10v}{v}, \quad d = \frac{1.708v + 0.404 - 1.481u}{v}$$

$$D_i = 16.518 + 1.481 \frac{c_r}{c_t} c_i - \frac{d_r}{d_t} d_i$$

$$u_i^a = \frac{10.872 + 0.404 \frac{c_r}{c_t} c_i - 4 \frac{d_r}{d_t} d_i}{D_i}, \quad v_i^a = \frac{5.520}{D_i}$$

A.12 UVW 色差与显色指数

$$W_i^* = 25Y_i^{\frac{1}{3}} - 17, \quad U_i^* = 13W_i^*(u_i^a - u_r), \quad V_i^* = 13W_i^*(v_i^a - v_r)$$

$$\Delta E_i = \sqrt{(U_i^* - U_{i,ref}^*)^2 + (V_i^* - V_{i,ref}^*)^2 + (W_i^* - W_{i,ref}^*)^2}$$

$$R_i = 100 - 4.6\Delta E_i$$

A.13 Ra、Rmin8 与最优比例筛选

$$R_a(p) = \frac{1}{8} \sum_{i=1}^8 R_i(p), \quad R_{min,8}(p) = \min_{1 \leq i \leq 8} R_i(p)$$

$$p_{Ra} = \arg \max_{0 \leq p \leq 1} R_a(p), \quad p_{min} = \arg \max_{0 \leq p \leq 1} R_{min,8}(p)$$